

A CONVERGENT COUNTEREXAMPLE TO THE RADICAL JACOBIAN CONTAINMENT

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ABSTRACT. We construct a convergent power series f in eight variables such that $f \notin \mathfrak{m}\sqrt{\text{Jac}(f)}$ in $\mathbb{C}[[y_1, \dots, y_8]]$. The construction begins with a homogeneous cubic whose critical locus contains 56 distinguished lines. A quartic perturbation is corrected in orders at least five by preserving the corresponding critical values on an affine hyperplane. The resulting critical locus contains 56 smooth branches. Their radical ideal K satisfies $f \in K^{(2)} \setminus \mathfrak{m}K$. The noncontainment is detected in degree four; its proof uses explicit low-degree equations and a constant-coefficient differential operator.

1. INTRODUCTION AND STATEMENT

For $F \in \mathbb{C}[[y_1, \dots, y_n]]$, write $\partial_i = \partial/\partial y_i$ and $\text{Jac}(F) = (\partial_1 F, \dots, \partial_n F)$. Eisenbud and Mazur asked whether $F \in \mathfrak{m}\sqrt{\text{Jac}(F)}$ for every $F \in \mathfrak{m} = (y_1, \dots, y_n)$; see the problem on page 2 of the arXiv version of [2]. We construct a convergent counterexample in eight variables.

Set

$$B = \mathbb{C}[y_1, \dots, y_8], \quad R = \mathbb{C}[[y_1, \dots, y_8]], \quad \mathfrak{m} = (y_1, \dots, y_8)R,$$

and define

$$\begin{aligned} s &= \sum_{i=1}^8 y_i, & \ell &= \frac{s}{3}, & h &= y_1 y_2 y_3 y_4, \\ g &= \sum_{i=1}^8 y_i^3 - \frac{s}{2} \sum_{i=1}^8 y_i^2 + \frac{s^3}{18}. \end{aligned} \tag{1.1}$$

Let $\mathcal{S}$ be the set of three-element subsets of $\{1, \dots, 8\}$. For $S \in \mathcal{S}$, let $p_S \in \mathbb{C}^8$ be its indicator vector and put $e_S = \prod_{i \in S} y_i$. There are $\binom{8}{3} = 56$ such points, all lying on the affine hyperplane $H = \{\ell = 1\}$.

Theorem 1.1. *There exist convergent power series $a_S(z) \in z^2 \mathbb{C}\{z\}$ such that*

$$f(y) = g(y) + h(y) + \sum_{S \in \mathcal{S}} a_S(\ell(y)) e_S(y) \tag{1.2}$$

satisfies

$$f \notin \mathfrak{m}\sqrt{\text{Jac}(f)} \quad \text{in } R.$$

In particular, $f \equiv g + h \pmod{\mathfrak{m}^5}$. More precisely, there are 56 distinct prime ideals $P_S \subset R$, each defining a smooth formal curve with tangent line $\mathbb{C}p_S$, such that, for $K = \bigcap_{S \in \mathcal{S}} P_S$,

$$R/P_S \cong \mathbb{C}[[z]], \quad \text{Jac}(f) \subseteq K, \quad f \in \bigcap_{S \in \mathcal{S}} P_S^2, \quad f \notin \mathfrak{m}K. \tag{1.3}$$

The series a_S are specified uniquely by the critical-value equations in Proposition 3.1. For a radical ideal $I \subset R$, we use the convention

$$I^{(2)} = \bigcap_{P \in \text{Min}(I)} (I^2 R_P \cap R).$$

Corollary 1.2. *The ideal K of Theorem 1.1 is radical, with exactly 56 minimal primes of height seven, and*

$$f \in K^{(2)} \setminus \mathfrak{m}K.$$

Eisenbud and Mazur relate their question to evolutions of reduced local algebras in [2, Corollary 2]. The symbolic-square formulation and its relation to evolutions are discussed in [1, §2.3]. The ideal constructed here is reducible. For each individual branch, $f \in P_S^2 \subseteq \mathfrak{m}P_S$. Thus the example does not give a prime ideal P with $P^{(2)} \not\subseteq \mathfrak{m}P$. Positive results for certain primes in formal power series rings appear in [3, Theorem 2.1].

The proof uses the following three preliminary facts.

- The homogeneous equations of the points p_S in degrees two and three are generated by $\partial_1 g, \dots, \partial_8 g$, and these generators have no syzygies with linear coefficients (Lemma 2.1). This follows from coefficient comparisons and interpolation by the monomials e_S .
- The quartic h vanishes at every p_S but does not belong to $\text{Jac}(g)$ (Lemma 2.2). An explicit linear functional separates it from the degree-four part of that ideal.
- The points p_S are nondegenerate critical points of $g|_H$ with value zero, and $e_T(p_S) = \delta_{TS}$ (Lemma 2.3). Their Hessians are restrictions of diagonal forms.

Sections 3–4 construct f and its critical branches. Section 5 proves the noncontainment by comparing degrees three and four. The longer coefficient calculation for Lemma 2.1 is deferred to the appendix.

All rings are ordinary commutative $\mathbb{C}$ -algebras. We write B_d for the degree- d part of B , $\partial_i = \partial/\partial y_i$, and $\mathbb{C}\{z\}$ for the ring of convergent power series at zero. All ideal containments in the proof are taken in R , unless an ideal of B is specified.

2. EQUATIONS AND CRITICAL POINTS OF THE CUBIC

Define homogeneous quadrics and a homogeneous ideal of B by

$$q_i = y_i^2 - \ell y_i, \quad J = (q_1, \dots, q_8).$$

For $d \geq 0$, put

$$J_d = J \cap B_d, \quad C_d = \{b \in B_d : b(p_S) = 0 \text{ for every } S \in \mathcal{S}\}.$$

Since $q_i(p_S) = 0$, we have $J_d \subseteq C_d$.

Lemma 2.1 (Low-degree equations). *In B , one has*

$$J = (\partial_1 g, \dots, \partial_8 g), \quad C_1 = 0, \quad C_2 = J_2, \quad C_3 = J_3.$$

Moreover, the linear map

$$B_1^{\oplus 8} \longrightarrow B_3, \quad (L_1, \dots, L_8) \longmapsto \sum_{i=1}^8 L_i \partial_i g$$

is injective.

The proof is given in Appendix A.

Lemma 2.2 (The quartic obstruction). *One has $h \in C_4$ and $h \notin J_4$.*

Proof. Every p_S has only three nonzero coordinates, so $h(p_S) = 0$. Let $I = (s, y_1^2, \dots, y_8^2) \subset B$; then $J \subseteq I$. Define a linear functional $\Lambda : B_4 \rightarrow \mathbb{C}$ by

$$\Lambda(b) = \left[\prod_{i=1}^4 (\partial_i - \partial_{i+4}) \right] b \Big|_{y=0}. \quad (2.1)$$

Each factor in this differential operator annihilates s . Consequently, $\Lambda(sB_3) = 0$ by the Leibniz rule. In every term of the expanded operator, each variable is differentiated at most once. Hence $\Lambda(y_i^2 B_2) = 0$ for every i . Thus $\Lambda(I_4) = 0$, whereas $\Lambda(h) = 1$. It follows that $h \notin J_4$. $\square$

Lemma 2.3 (Critical points and interpolation). *Each p_S is a nondegenerate critical point of $g|_H$, with critical value zero. Furthermore, $e_T(p_S) = \delta_{TS}$ for $S, T \in \mathcal{S}$.*

Proof. On H , where $s = 3$, the restriction of g is

$$g|_H = \sum_{i=1}^8 y_i^3 - \frac{3}{2} \sum_{i=1}^8 y_i^2 + \frac{3}{2}.$$

Its value and differential vanish at every p_S . Let

$$V = \ker \ell = \left\{ v \in \mathbb{C}^8 : \sum_{i=1}^8 v_i = 0 \right\}.$$

The Hessian on $V = T_{p_S} H$ is the restriction of $\text{diag}(d_1, \dots, d_8)$, where $d_i = 6(p_S)_i - 3$. There are three entries equal to 3 and five equal to -3 . If $v \in V$ is in the radical of this restricted bilinear form, then $d_i v_i = c$ for some $c \in \mathbb{C}$ and every i . Therefore

$$0 = \sum_i v_i = c \sum_i d_i^{-1} = -\frac{2}{3}c,$$

so $v = 0$. This proves nondegeneracy. Finally, $e_T(p_S) = 1$ precisely when $T \subseteq S$; since both sets have cardinality three, this is equivalent to $T = S$. $\square$

3. A DEFORMATION WITH VANISHING CRITICAL VALUES

Introduce independent parameters z and $a = (a_T)_{T \in \mathcal{S}}$, and let

$$G(z, u, a) = g(u) + zh(u) + \sum_{T \in \mathcal{S}} a_T e_T(u), \quad u \in H. \quad (3.1)$$

The notation $d_u G$ below means the differential along the affine hyperplane H .

Proposition 3.1. *For every $S \in \mathcal{S}$, there is a unique holomorphic germ $u_S(z, a)$ with values in H such that*

$$u_S(0, 0) = p_S, \quad d_u G(z, u_S(z, a), a) = 0.$$

Set $c_S(z, a) = G(z, u_S(z, a), a)$. There is a unique holomorphic germ $a(z) = (a_T(z))_{T \in \mathcal{S}}$ with $a(0) = 0$ such that

$$c_S(z, a(z)) = 0 \quad (S \in \mathcal{S}). \quad (3.2)$$

Every component satisfies $a_T(z) \in z^2 \mathbb{C}\{z\}$.

Proof. (1) Choose affine coordinates on H near p_S . The critical-point equations for G are seven holomorphic equations. Their Jacobian with respect to the seven coordinates of u at $(z, u, a) = (0, p_S, 0)$ is the Hessian of $g|_H$ at p_S . It is invertible by Lemma 2.3. The holomorphic implicit function theorem gives the asserted unique germ $u_S(z, a)$.

(2) The critical values c_S are holomorphic germs and $c_S(0, 0) = 0$. In differentiating c_S , the terms involving derivatives of u_S vanish because $d_u G = 0$. At $(z, a) = (0, 0)$, this gives

$$\frac{\partial c_S}{\partial a_T}(0, 0) = e_T(p_S) = \delta_{TS}, \quad \frac{\partial c_S}{\partial z}(0, 0) = h(p_S) = 0.$$

The Jacobian of $(c_S)_{S \in \mathcal{S}}$ with respect to a is therefore the identity. A second application of the holomorphic implicit function theorem gives the unique germ $a(z)$ satisfying (3.2).

Differentiating those equations at zero yields $a'_T(0) = 0$ for every T .

This proves $a_T(z) \in z^2 \mathbb{C}\{z\}$. $\square$

Use these series in (1.2), and set

$$\tilde{G}(z, u) = G(z, u, a(z)), \quad v_S(z) = u_S(z, a(z)).$$

Since $\ell(zu) = z$ for $u \in H$, homogeneity gives the identity

$$f(zu) = z^3 \tilde{G}(z, u). \quad (3.3)$$

The series f converges near the origin. Moreover, $a_S(\ell)e_S \in \mathfrak{m}^5$, so its terms of degrees three and four are exactly g and h .

Remark 3.2. Both implicit-function problems have invertible Jacobians at the base point, so they also have unique formal solutions. These are the Taylor series of the holomorphic germs constructed above. The only analytic existence input is the holomorphic implicit function theorem.

4. THE CRITICAL BRANCHES AND THEIR EQUATIONS

For $S \in \mathcal{S}$, define

$$\gamma_S(z) = zv_S(z).$$

Then

$$\ell(\gamma_S(z)) = z, \quad \gamma_S(z) = zp_S + O(z^2). \quad (4.1)$$

Proposition 4.1. *For every $S \in \mathcal{S}$,*

$$f(\gamma_S(z)) = 0, \quad (\partial_i f)(\gamma_S(z)) = 0 \quad (1 \leq i \leq 8).$$

The ideal

$$P_S = \ker(R \longrightarrow \mathbb{C}[[z]], \quad y_i \longmapsto \gamma_{S,i}(z)) \quad (4.2)$$

defines a smooth formal curve with tangent line $\mathbb{C}p_S$, and $R/P_S \cong \mathbb{C}[[z]]$. These 56 primes are distinct, and

$$f \in P_S^2 \quad (S \in \mathcal{S}).$$

Proof. (1) By (3.2) and (3.3), $f(\gamma_S(z)) = 0$. For $w \in V = \ker \ell$, differentiation of (3.3) in the u -direction w , followed by substitution $u = v_S(z)$, gives

$$z df_{\gamma_S(z)}(w) = z^3 d_u \tilde{G}(z, v_S(z))(w) = 0.$$

Cancellation of z in $\mathbb{C}[[z]]$ shows that $df_{\gamma_S(z)}$ vanishes on V . Differentiating $f(\gamma_S(z)) = 0$ also gives

$$df_{\gamma_S(z)}(\gamma'_S(z)) = 0.$$

Since $\ell(\gamma'_S(z)) = 1$, the constant vectors in V , together with $\gamma'_S(z)$, span $\mathbb{C}[[z]]^8$ over $\mathbb{C}[[z]]$. Thus every partial derivative of f vanishes along γ_S .

- (2) The evaluation map in (4.2) is surjective: it sends ℓ to z , and hence sends $A(\ell)$ to $A(z)$ for every $A(z) \in \mathbb{C}[[z]]$. Its kernel P_S is prime and $R/P_S \cong \mathbb{C}[[z]]$. The tangent vector in (4.1) is $p_S \neq 0$. The lines $\mathbb{C}p_S$ are pairwise distinct, since $\ell(p_S) = 1$ for every S . Hence the primes are pairwise distinct.
- (3) To prove the square containment, use the linear coordinates $t = \ell$ and $x_j = y_j$ for $1 \leq j \leq 7$ on $\mathbb{C}^8$. Set

$$w_j = x_j - \gamma_{S,j}(t) \quad (1 \leq j \leq 7).$$

These give formal coordinates with

$$R = \mathbb{C}[[t, w_1, \dots, w_7]], \quad P_S = (w_1, \dots, w_7).$$

Modulo P_S^2 , write

$$f = A(t) + \sum_{j=1}^7 B_j(t)w_j.$$

Restriction to the branch gives $A(t) = 0$. The vanishing of all partial derivatives along the branch, which is preserved by the coordinate change by the chain rule, gives $B_j(t) = 0$ for every j . Thus $f \in P_S^2$. $\square$

Let $K = \bigcap_{S \in \mathcal{S}} P_S$. It is radical, and Proposition 4.1 gives

$$\text{Jac}(f) \subseteq K, \quad \sqrt{\text{Jac}(f)} \subseteq K. \quad (4.3)$$

The next lemma relates the equations of the deformed branches to the low-degree equations of their tangent directions.

Lemma 4.2. *One has*

$$K \subseteq \mathfrak{m}^2, \quad K \subseteq \text{Jac}(f) + \mathfrak{m}^4, \quad \mathfrak{m}K \subseteq \mathfrak{m} \text{Jac}(f) + \mathfrak{m}^5.$$

Proof. Let $b \in K$ be nonzero, and let b_d be its first nonzero homogeneous part. Substitution of (4.1) gives

$$0 = b(\gamma_S(z)) = z^d b_d(p_S) + O(z^{d+1}),$$

so $b_d \in C_d$. All branches pass through the origin, so $K \subseteq \mathfrak{m}$. Since $C_1 = 0$, it follows that $K \subseteq \mathfrak{m}^2$.

For arbitrary $b \in K$, its quadratic part lies in $C_2 = J_2$. Subtract a constant linear combination of the $\partial_i f$ to cancel that part. The remainder belongs to $K \cap \mathfrak{m}^3$, because $\text{Jac}(f) \subseteq K$, and its cubic part lies in $C_3 = J_3$. Subtract a combination of the $\partial_i f$ with linear coefficients to cancel the cubic part. The final remainder belongs to $\mathfrak{m}^4$. Here the leading quadratic parts of $\partial_i f$ are $\partial_i g$, and $J = (\partial_1 g, \dots, \partial_8 g)$ by Lemma 2.1. This proves $K \subseteq \text{Jac}(f) + \mathfrak{m}^4$; multiplication by $\mathfrak{m}$ gives the last containment. $\square$

5. THE DEGREE-FOUR OBSTRUCTION

Proof of Theorem 1.1. Propositions 3.1 and 4.1 give convergence and all the asserted branch properties. Suppose $f \in \mathfrak{m}K$. By Lemma 4.2, $f \in \mathfrak{m} \text{Jac}(f) + \mathfrak{m}^5$. Since each $\partial_i f$ has order two, truncating the coefficients gives $L_i \in B_1$ and $Q_i \in B_2$ such that

$$f \equiv \sum_{i=1}^8 (L_i + Q_i) \partial_i f \pmod{\mathfrak{m}^5}. \quad (5.1)$$

(1) Comparing degree three in (5.1) gives

$$g = \sum_i L_i \partial_i g.$$

Euler's identity is $g = \frac{1}{3} \sum_i y_i \partial_i g$. By the injectivity assertion of Lemma 2.1,

$$L_i = \frac{y_i}{3} \quad (1 \leq i \leq 8).$$

(2) Since $\partial_i f \equiv \partial_i g + \partial_i h \pmod{\mathfrak{m}^4}$, comparison in degree four now yields

$$h = \frac{1}{3} \sum_i y_i \partial_i h + \sum_i Q_i \partial_i g = \frac{4}{3} h + \sum_i Q_i \partial_i g.$$

Hence $-\frac{1}{3}h \in J_4$, contradicting Lemma 2.2.

Therefore $f \notin \mathfrak{m}K$. Finally, (4.3) gives $\mathfrak{m}\sqrt{\text{Jac}(f)} \subseteq \mathfrak{m}K$, and thus $f \notin \mathfrak{m}\sqrt{\text{Jac}(f)}$. $\square$

Proof of Corollary 1.2. In the coordinates used in Proposition 4.1, each P_S is generated by seven of the eight parameters. Thus it has height seven. The distinct primes P_S have the same height, so none contains another. They are exactly the minimal primes of their intersection K . Localizing at P_S gives $KR_{P_S} = P_SR_{P_S}$. Since $f \in P_S^2$ for every S , we obtain

$$f \in \bigcap_{S \in \mathcal{S}} (K^2 R_{P_S} \cap R) = K^{(2)}.$$

The nonmembership $f \notin \mathfrak{m}K$ was proved above. $\square$

Remark 5.1. The noncontainment also holds in the convergent power series ring $\mathbb{C}\{y_1, \dots, y_8\}$. Indeed, a representation of f as an element of the maximal ideal times the radical of its analytic Jacobian ideal would give such a representation in R after taking Taylor series.

APPENDIX A. PROOF OF THE LOW-DEGREE EQUATIONS

Proof of Lemma 2.1. (1) *The Jacobian generators.* Direct differentiation of (1.1) gives

$$\partial_i g = 3q_i - \frac{1}{2} \sum_{j=1}^8 q_j.$$

If $\mathbf{1} = (1, \dots, 1)^\top$, the change-of-generators matrix is $3I_8 - \frac{1}{2}\mathbf{1}\mathbf{1}^\top$. Its eigenvalues are 3 on $\ker \ell$ and -1 on $\mathbb{C}\mathbf{1}$. It is invertible, proving $J = (\partial_1 g, \dots, \partial_8 g)$.

- (2) *Linear equations.* Let ε_i denote the i th standard basis vector of $\mathbb{C}^8$. For distinct i, j , choose a two-element subset $T \subset \{1, \dots, 8\} \setminus \{i, j\}$. Then

$$p_{T \cup \{i\}} - p_{T \cup \{j\}} = \varepsilon_i - \varepsilon_j.$$

These differences span $\ker \ell$. Since $\ell(p_S) = 1$, the vectors p_S span $\mathbb{C}^8$. Therefore $C_1 = 0$.

- (3) *Quadratic equations.* The coefficient of y_i^2 in q_i is $2/3$, and no other square appears in q_i . Given $b \in C_2$, subtract a constant linear combination of the q_i to obtain

$$b' = \sum_{i < j} b_{ij} y_i y_j \in C_2.$$

Evaluation on a triple gives

$$b_{ij} + b_{ik} + b_{jk} = 0 \quad (i, j, k \text{ distinct}), \quad (\text{A.1})$$

where $b_{ij} = b_{ji}$. Fix distinct i, j and set $d_k = b_{ik} - b_{jk}$ for $k \notin \{i, j\}$. Subtracting the equations for $\{i, k, l\}$ and $\{j, k, l\}$ gives $d_k + d_l = 0$ whenever $k, l \notin \{i, j\}$ are distinct. There are at least three such indices, so $d_k = 0$ for every k . Thus, for fixed k , all b_{ik} with $i \neq k$ are equal. Symmetry then makes all b_{ij} equal, and (A.1) makes their common value zero. Hence $b' = 0$ and $C_2 = J_2$.

- (4) *Linear syzygies.* It suffices to treat the generators $3q_i = 3y_i^2 - sy_i$. Suppose

$$\sum_{i,j=1}^8 a_{ij} y_j (3y_i^2 - sy_i) = 0.$$

The coefficient of y_i^3 is $2a_{ii}$, so $a_{ii} = 0$. For $i \neq j$, the coefficient of $y_i^2 y_j$ is now $2a_{ij} - a_{ji}$. Interchanging i, j gives both $2a_{ij} = a_{ji}$ and $2a_{ji} = a_{ij}$, whence $a_{ij} = a_{ji} = 0$. Thus the quadrics q_i , and therefore the partial derivatives of g , have no nonzero syzygies with linear coefficients.

(5) *Cubic equations.* The evaluation map

$$B_3 \longrightarrow \mathbb{C}^{\mathcal{S}}, \quad b \longmapsto (b(p_S))_{S \in \mathcal{S}}$$

is surjective because $e_T(p_S) = \delta_{TS}$. Therefore

$$\dim_{\mathbb{C}} C_3 = \binom{10}{3} - 56 = 64.$$

The injectivity proved in the preceding step gives $\dim_{\mathbb{C}} J_3 = 8 \dim_{\mathbb{C}} B_1 = 64$. Since $J_3 \subseteq C_3$, equality follows.

This proves all the assertions. □

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